

Assignment 15: Network & System Tools + Compression & Text Processing

Part 1: User & Group Permissions

Task 1.1: Create Users & Groups

Commands Used:

```
sudo groupadd network_team
sudo useradd -m -G network_team alice
sudo useradd -m -G network_team bob
```

```
ubuntu@ip-172-31-23-124:~$ sudo groupadd network_team
sudo useradd -m -G network_team alice
sudo useradd -m -G network_team bob
ubuntu@ip-172-31-23-124:~$ |
```

Explanation:

Created a group `network_team`, then added users `alice` and `bob` to this group with home directories.

Task 1.2: Set Directory Permissions

Commands Used:

```
sudo mkdir /opt/network_data
sudo chown root:network_team /opt/network_data
sudo chmod 770 /opt/network_data
```

```
ubuntu@ip-172-31-23-124:~$ sudo mkdir /opt/network_data
sudo chown root:network_team /opt/network_data
sudo chmod 770 /opt/network_data
ubuntu@ip-172-31-23-124:~$ |
```

Explanation:

Created a shared directory and granted full permissions to the group `network_team`.

View Permissions:

Commands Used:

```
ls -ld /opt/network_data
```

```
ubuntu@ip-172-31-23-124:~$ ls -ld /opt/network_data
drwxrwx--- 2 root network_team 4096 May 16 22:27 /opt/network_data
```

Part 2: Network Tools & Real:Time Checks

Task 2.1: Connectivity to google.com

Commands Used:

ping -c 4 google.com

```
ubuntu@ip-172-31-23-124:~$ ping -c 4 google.com
PING google.com (172.253.115.113) 56(84) bytes of data.
64 bytes from bg-in-f113.1e100.net (172.253.115.113): icmp_seq=1 ttl=106 time=1.52 ms
64 bytes from bg-in-f113.1e100.net (172.253.115.113): icmp_seq=2 ttl=106 time=2.23 ms
64 bytes from bg-in-f113.1e100.net (172.253.115.113): icmp_seq=3 ttl=106 time=1.73 ms
64 bytes from bg-in-f113.1e100.net (172.253.115.113): icmp_seq=4 ttl=106 time=1.57 ms

--- google.com ping statistics ---
4 packets transmitted, 4 received, 0% packet loss, time 3005ms
rtt min/avg/max/mdev = 1.521/1.764/2.233/0.281 ms
```

traceroute google.com

```
ubuntu@ip-172-31-23-124:~$ traceroute google.com
traceroute to google.com (172.253.115.102), 30 hops max, 60 byte packets
 1 100.100.38.108 (100.100.38.108) 0.806 ms 100.100.32.86 (100.100.32.86) 0.758 ms 100.100.6.24 (100.100.6.24) 0.682
ms
 2 240.0.184.33 (240.0.184.33) 1.004 ms 240.0.184.12 (240.0.184.12) 0.898 ms 240.0.236.15 (240.0.236.15) 0.939 ms
 3 100.100.2.32 (100.100.2.32) 1.593 ms 100.100.2.122 (100.100.2.122) 1.613 ms 240.0.184.7 (240.0.184.7) 1.569 ms
 4 100.100.4.28 (100.100.4.28) 0.951 ms * *
 5 * * 99.83.65.3 (99.83.65.3) 3.772 ms
 6 * 142.251.52.62 (142.251.52.62) 1.778 ms *
 7 192.178.242.24 (192.178.242.24) 1.864 ms 192.178.243.4 (192.178.243.4) 1.434 ms 142.251.67.234 (142.251.67.234) 1
.406 ms
 8 142.251.49.17 (142.251.49.17) 1.921 ms 216.239.48.101 (216.239.48.101) 1.973 ms 192.178.110.112 (192.178.110.112)
1.857 ms
 9 142.251.49.162 (142.251.49.162) 5.111 ms 216.239.49.197 (216.239.49.197) 2.723 ms 142.250.211.189 (142.250.211.189
) 2.145 ms
10 142.250.59.233 (142.250.59.233) 2.666 ms 172.253.66.58 (172.253.66.58) 2.643 ms 172.253.79.84 (172.253.79.84) 6.2
67 ms
11 172.253.67.67 (172.253.67.67) 2.594 ms 172.253.66.197 (172.253.66.197) 1.900 ms 108.170.230.5 (108.170.230.5) 1.7
57 ms
12 * * *
13 * * *
14 * * *
15 * * *
16 * * *
17 * * *
18 * * *
19 * * *
20 * * *
21 bg-in-f102.1e100.net (172.253.115.102) 1.880 ms 1.862 ms 1.826 ms
```

mtr --report google.com

```
ubuntu@ip-172-31-23-124:~$ mtr --report google.com
Start: 2025-05-16T22:34:28+0000
HOST: ip-172-31-23-124
 1. |-- 100.100.6.80          0.0% 10 4.0 3.1 0.6 17.4 5.1
 2. |-- 240.0.184.34         0.0% 10 1.3 1.1 1.0 1.4 0.1
 3. |-- 100.100.36.96        0.0% 10 1.7 5.0 1.0 35.3 10.6
 4. |-- 99.82.180.135        0.0% 10 1.8 1.7 1.5 2.2 0.2
 5. |-- 142.251.71.127       0.0% 10 2.4 2.2 1.8 2.4 0.2
 6. |-- 192.178.243.4        0.0% 10 16.8 3.3 1.5 16.8 4.8
 7. |-- 216.239.50.97        0.0% 10 2.2 2.5 2.2 2.9 0.2
 8. |-- 192.178.80.183       0.0% 10 3.3 4.0 3.3 4.5 0.4
 9. |-- 142.251.251.15       0.0% 10 2.5 3.1 2.4 5.6 0.9
10. |-- 192.178.75.147       0.0% 10 2.6 3.1 2.6 4.0 0.5
11. |-- ???                 100.0 10 0.0 0.0 0.0 0.0 0.0
12. |-- ???                 100.0 10 0.0 0.0 0.0 0.0 0.0
13. |-- ???                 100.0 10 0.0 0.0 0.0 0.0 0.0
14. |-- ???                 100.0 10 0.0 0.0 0.0 0.0 0.0
15. |-- ???                 100.0 10 0.0 0.0 0.0 0.0 0.0
16. |-- ???                 100.0 10 0.0 0.0 0.0 0.0 0.0
17. |-- ???                 100.0 10 0.0 0.0 0.0 0.0 0.0
18. |-- ???                 100.0 10 0.0 0.0 0.0 0.0 0.0
19. |-- ???                 100.0 10 0.0 0.0 0.0 0.0 0.0
20. |-- on-in-f101.1e100.net 0.0% 10 2.3 2.3 2.2 2.9 0.2
ubuntu@ip-172-31-23-124:~$
```

Explanation:

Tested internet connectivity using ping, traced the route with traceroute, and analyzed hops with mtr.

Task 2.2: Open Ports & Listening Services

Commands Used:

```
sudo netstat -tuln
```

```
sudo ss -tulwn
```

```
ubuntu@ip-172-31-23-124:~$ sudo netstat -tuln
Active Internet connections (only servers)
Proto Recv-Q Send-Q Local Address           Foreign Address         State
tcp        0      0 127.0.0.53:53          0.0.0.0:*               LISTEN
tcp        0      0 127.0.0.54:53          0.0.0.0:*               LISTEN
tcp6       0      0 :::22                  :::*                     LISTEN
udp        0      0 127.0.0.54:53          0.0.0.0:*
udp        0      0 127.0.0.53:53          0.0.0.0:*
udp        0      0 172.31.23.124:68       0.0.0.0:*
udp        0      0 127.0.0.1:323          0.0.0.0:*
udp6       0      0 :::1:323               :::*

ubuntu@ip-172-31-23-124:~$ sudo ss -tulwn
Netid      State      Recv-Q     Send-Q     Local Address:Port      Peer Address:Port      Process
icmp6      UNCONN    0           0           *%enX0:58               *:*
udp        UNCONN    0           0           127.0.0.54:53           0.0.0.0:*
udp        UNCONN    0           0           127.0.0.53%lo:53        0.0.0.0:*
udp        UNCONN    0           0           172.31.23.124%enX0:68    0.0.0.0:*
udp        UNCONN    0           0           127.0.0.1:323           0.0.0.0:*
udp        UNCONN    0           0           [::1]:323               [::]:*
tcp        LISTEN    0           4096        127.0.0.53%lo:53        0.0.0.0:*
tcp        LISTEN    0           4096        127.0.0.54:53           0.0.0.0:*
tcp        LISTEN    0           4096        *:22                     *:*
ubuntu@ip-172-31-23-124:~$ |
```

Explanation:

Displayed all open TCP/UDP ports and listening services on the system.

Task 2.3: Test Remote Port 443

Commands Used:

```
telnet google.com 443
```

```
nc -zv google.com 443
```

```
ubuntu@ip-172-31-23-124:~$ telnet google.com 443
Trying 64.233.180.139...
Connected to google.com.
Escape character is '^]'.
Connection closed by foreign host.
ubuntu@ip-172-31-23-124:~$ nc -zv google.com 443
Connection to google.com (64.233.180.101) 443 port [tcp/https] succeeded!
ubuntu@ip-172-31-23-124:~$ |
```

Explanation:

Tested connectivity to Google on port 443 to verify HTTPS access.

Task 2.4: Check Network Interfaces

Commands Used:

ifconfig
ip addr

```
ubuntu@ip-172-31-23-124:~$ ifconfig
enX0: flags=4163<UP,BROADCAST,RUNNING,MULTICAST> mtu 9001
    inet 172.31.23.124 netmask 255.255.240.0 broadcast 172.31.31.255
    inet6 fe80::8ff:ccff:feaa:e7e5 prefixlen 64 scopeid 0x20<link>
    ether 0a:ff:cc:aa:e7:e5 txqueuelen 1000 (Ethernet)
    RX packets 54078 bytes 73685606 (73.6 MB)
    RX errors 0 dropped 0 overruns 0 frame 0
    TX packets 5863 bytes 566481 (566.4 KB)
    TX errors 0 dropped 0 overruns 0 carrier 0 collisions 0

lo: flags=73<UP,LOOPBACK,RUNNING> mtu 65536
    inet 127.0.0.1 netmask 255.0.0.0
    inet6 ::1 prefixlen 128 scopeid 0x10<host>
    loop txqueuelen 1000 (Local Loopback)
    RX packets 358 bytes 39613 (39.6 KB)
    RX errors 0 dropped 0 overruns 0 frame 0
    TX packets 358 bytes 39613 (39.6 KB)
    TX errors 0 dropped 0 overruns 0 carrier 0 collisions 0

ubuntu@ip-172-31-23-124:~$ ip addr
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group default qlen 1000
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host lo
        valid_lft forever preferred_lft forever
    inet6 ::1/128 scope host noprefixroute
        valid_lft forever preferred_lft forever
2: enX0: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 9001 qdisc fq_codel state UP group default qlen 1000
    link/ether 0a:ff:cc:aa:e7:e5 brd ff:ff:ff:ff:ff:ff
    inet 172.31.23.124/20 metric 100 brd 172.31.31.255 scope global dynamic enX0
        valid_lft 2804sec preferred_lft 2804sec
    inet6 fe80::8ff:ccff:feaa:e7e5/64 scope link
        valid_lft forever preferred_lft forever
ubuntu@ip-172-31-23-124:~$
```

Explanation:

Listed network interfaces and their IP configurations.

Task 2.5: DNS Lookup

Commands Used:

nslookup google.com

```
ubuntu@ip-172-31-23-124:~$ nslookup google.com
Server:      127.0.0.53
Address:     127.0.0.53#53

Non-authoritative answer:
Name:   google.com
Address: 142.251.179.139
Name:   google.com
Address: 142.251.179.101
Name:   google.com
Address: 142.251.179.100
Name:   google.com
Address: 142.251.179.102
Name:   google.com
Address: 142.251.179.138
Name:   google.com
Address: 142.251.179.113
Name:   google.com
Address: 2607:f8b0:4004:c08::71
Name:   google.com
Address: 2607:f8b0:4004:c08::64
Name:   google.com
Address: 2607:f8b0:4004:c08::8b
Name:   google.com
Address: 2607:f8b0:4004:c08::65
```

dig google.com

```
ubuntu@ip-172-31-23-124:~$ dig google.com

; <<>> DiG 9.18.30-0ubuntu0.24.04.2-Ubuntu <<>> google.com
;; global options: +cmd
;; Got answer:
;; ->>HEADER<<- opcode: QUERY, status: NOERROR, id: 51949
;; flags: qr rd ra; QUERY: 1, ANSWER: 6, AUTHORITY: 0, ADDITIONAL: 1

;; OPT PSEUDOSECTION:
; EDNS: version: 0, flags:; udp: 65494
;; QUESTION SECTION:
;google.com.                IN      A

;; ANSWER SECTION:
google.com.                223     IN      A      142.251.179.100
google.com.                223     IN      A      142.251.179.113
google.com.                223     IN      A      142.251.179.139
google.com.                223     IN      A      142.251.179.101
google.com.                223     IN      A      142.251.179.102
google.com.                223     IN      A      142.251.179.138

;; Query time: 0 msec
;; SERVER: 127.0.0.53#53(127.0.0.53) (UDP)
;; WHEN: Fri May 16 22:38:40 UTC 2025
;; MSG SIZE rcvd: 135
```

Explanation:

Verified DNS resolution for google.com using two methods.

Task 2.6: Download Test File

Commands Used:

```
wget https://example.com/testfile.txt
curl -O https://example.com/testfile.txt
```

```
ubuntu@ip-172-31-23-124:~$ wget https://example.com/testfile.txt
--2025-05-16 22:40:40-- https://example.com/testfile.txt
Resolving example.com (example.com)... 96.7.128.198, 23.192.228.80, ...
Connecting to example.com (example.com)|96.7.128.198|:443... connected.
HTTP request sent, awaiting response... 404 Not Found
2025-05-16 22:40:41 ERROR 404: Not Found.

ubuntu@ip-172-31-23-124:~$ curl -O https://example.com/testfile.txt
  % Total    % Received % Xferd  Average Speed   Time    Time     Time  Current
                                 Dload  Upload  Total   Spent    Left   Speed
100 1256 100 1256  0     0    3112      0 --:--:-- --:--:-- --:--:-- 3124
ubuntu@ip-172-31-23-124:~$
```

Explanation:

Tested file downloads using wget and curl to ensure outbound HTTP access.

Task 2.7: Monitor Bandwidth in Real Time

Commands Used:

```
sudo iftop -i eth0
ip link show
sudo iftop -i enX0
```

```
ubuntu@ip-172-31-23-124:~$ sudo iftop -i eth0
interface: eth0
Error getting hardware address for interface: eth0
ioctl(SIOCGIFHWADDR): No such device
pcap_open_live(eth0): eth0: No such device exists (No such device exists)
ubuntu@ip-172-31-23-124:~$ ip link show
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN mode DEFAULT group default qlen 1000
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
2: enX0: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 9001 qdisc fq_codel state UP mode DEFAULT group default qlen 1000
    link/ether 0a:ff:cc:aa:e7:e5 brd ff:ff:ff:ff:ff:ff
ubuntu@ip-172-31-23-124:~$ sudo iftop -i enX0
interface: enX0
IP address is: 172.31.23.124
MAC address is: 0a:ff:cc:aa:e7:e5
```

	12.5Kb	25.0Kb	37.5Kb	50.0Kb	62.5Kb
ip-172-31-23-124.ec2.internal	=>	154.192.30.79		2.08Kb	1.66Kb 5.87Kb
	<=			160b	160b 4.22Kb
ip-172-31-23-124.ec2.internal	=>	ip-172-31-0-2.ec2.internal		0b	448b 358b
	<=			0b	477b 438b
ip-172-31-23-124.ec2.internal	=>	169.254.169.123		0b	61b 34b
	<=			0b	61b 34b
ip-172-31-23-124.ec2.internal	=>	b09-07.sysnet.ucsd.edu		0b	61b 34b
	<=			0b	61b 34b
ip-172-31-23-124.ec2.internal	=>	prod-ntp-3.ntp4.ps5.canonical.com		304b	61b 34b
	<=			304b	61b 34b
ip-172-31-23-124.ec2.internal	=>	208.67.75.242		0b	61b 34b
	<=			0b	61b 34b

TX: [REDACTED]	cum:	14.3KB	peak:	24.5Kb	rates:	2.38Kb	2.34Kb	6.35Kb
RX: [REDACTED]		10.8KB		25.3Kb		464b	880b	4.78Kb
TOTAL:		25.0KB		49.8Kb		2.83Kb	3.20Kb	11.1Kb

```
sudo nload enX0
```

```
Device enX0 [172.31.23.124] (1/1):
=====
Incoming:

Curr: 728.00 Bit/s
Avg: 3.27 kBit/s
Min: 360.00 Bit/s
Max: 25.48 kBit/s
Ttl: 78.60 MByte

Outgoing:

Curr: 7.34 kBit/s
Avg: 10.25 kBit/s
Min: 6.63 kBit/s
Max: 33.27 kBit/s
Ttl: 936.52 kByte
```

Explanation:

Used iftop and nload to monitor network traffic and bandwidth usage live.

Part 3: Compression & Decompression

Task 3.1: Archive Directory

Command:

```
sudo tar cvf network_data.tar /opt/network_data  
sudo tar cvf network_data.tar --absolute-names /opt/network_data
```

```
ubuntu@ip-172-31-23-124:~$ sudo tar cvf network_data.tar /opt/network_data  
tar: Removing leading '/' from member names  
/opt/network_data/  
ubuntu@ip-172-31-23-124:~$ sudo tar cvf network_data.tar --absolute-names /opt/network_data  
/opt/network_data/
```

Explanation:

Archived the shared directory into a .tar file.

Task 3.2: Compress Archive with gzip

Command:

```
gzip network_data.tar
```

```
ubuntu@ip-172-31-23-124:~$ sudo gzip network_data.tar  
ubuntu@ip-172-31-23-124:~$ |
```

Explanation:

Compressed the tar archive using gzip, creating network_data.tar.gz.

Task 3.3: Decompress gzip

Command:

```
gunzip network_data.tar.gz
```

```
ubuntu@ip-172-31-23-124:~$ gunzip network_data.tar.gz  
ubuntu@ip-172-31-23-124:~$ |
```

Explanation:

Decompressed the gzip archive to restore the .tar file.

Task 3.4: Compress & Decompress with bzip2

Commands:

```
bzip2 network_data.tar  
bunzip2 network_data.tar.bz2
```



```
ubuntu@ip-172-31-23-124:~$ bzip2 network_data.tar
ubuntu@ip-172-31-23-124:~$ bunzip2 network_data.tar.bz2
ubuntu@ip-172-31-23-124:~$ |
```

Explanation:

Compressed the tar file using bzip2 and then decompressed it.

View Compressed File Sizes:

Command:

ls -lh

```
ubuntu@ip-172-31-23-124:~$ ls -lh
total 24K
-rw-rw-r-- 1 ubuntu ubuntu 267 May 10 07:20 auth.log
-rw-rw-r-- 1 ubuntu ubuntu 10K May 16 22:49 network_data.tar
-rw-rw-r-- 1 ubuntu ubuntu 252 May 10 07:18 syslog.txt
-rw-rw-r-- 1 ubuntu ubuntu 1.3K May 16 22:41 testfile.txt
ubuntu@ip-172-31-23-124:~$ |
```

Explanation:

Displayed human:readable file sizes of original and compressed files.

Part 4: Text Processing with grep & awk

Task 4.1: Search for "error" in Log Files

Command:

```
grep "error" /var/log/syslog
```

```
ubuntu@ip-172-31-23-124:~$ grep "error" /var/log/syslog
2025-05-10T06:29:43.849584+00:00 ip-172-31-23-124 systemd[1]: apport-autoreport.path - Process error reports when automa
tic reporting is enabled (file watch) was skipped because of an unmet condition check (ConditionPathExists=/var/lib/appo
rt/autoreport).
2025-05-10T06:29:43.849590+00:00 ip-172-31-23-124 systemd[1]: apport-autoreport.timer - Process error reports when autom
atic reporting is enabled (timer based) was skipped because of an unmet condition check (ConditionPathExists=/var/lib/ap
port/autoreport).
2025-05-10T06:29:50.220892+00:00 ip-172-31-23-124 amazon-ssm-agent.amazon-ssm-agent[947]: 2025-05-10 06:29:48.5981 ERROR
EC2RoleProvider Failed to connect to Systems Manager with SSM role credentials. error calling RequestManagedInstanceRol
eToken: AccessDeniedException: Systems Manager's instance management role is not configured for account: 864981713828
2025-05-10T06:29:50.321227+00:00 ip-172-31-23-124 amazon-ssm-agent.amazon-ssm-agent[947]: 2025-05-10 06:29:48.5981 ERROR
[CredentialRefresher] Retrieve credentials produced error: no valid credentials could be retrieved for ec2 identity. De
fault Host Management Err: error calling RequestManagedInstanceRoleToken: AccessDeniedException: Systems Manager's insta
nce management role is not configured for account: 864981713828
2025-05-10T06:35:16.726805+00:00 ip-172-31-23-124 systemd[1]: apport-autoreport.path - Process error reports when automa
tic reporting is enabled (file watch) was skipped because of an unmet condition check (ConditionPathExists=/var/lib/appo
rt/autoreport).
2025-05-10T06:35:16.729062+00:00 ip-172-31-23-124 systemd[1]: apport-autoreport.timer - Process error reports when autom
atic reporting is enabled (timer based) was skipped because of an unmet condition check (ConditionPathExists=/var/lib/ap
port/autoreport).
2025-05-10T06:59:22.704685+00:00 ip-172-31-23-124 amazon-ssm-agent.amazon-ssm-agent[947]: 2025-05-10 06:59:22.6502 ERROR
EC2RoleProvider Failed to connect to Systems Manager with SSM role credentials. error calling RequestManagedInstanceRol
eToken: AccessDeniedException: Systems Manager's instance management role is not configured for account: 864981713828
2025-05-10T06:59:22.806211+00:00 ip-172-31-23-124 amazon-ssm-agent.amazon-ssm-agent[947]: 2025-05-10 06:59:22.6503 ERROR
[CredentialRefresher] Retrieve credentials produced error: no valid credentials could be retrieved for ec2 identity. De
fault Host Management Err: error calling RequestManagedInstanceRoleToken: AccessDeniedException: Systems Manager's insta
nce management role is not configured for account: 864981713828
2025-05-10T07:27:36.775454+00:00 ip-172-31-23-124 amazon-ssm-agent.amazon-ssm-agent[947]: 2025-05-10 07:27:36.7229 ERROR
EC2RoleProvider Failed to connect to Systems Manager with SSM role credentials. error calling RequestManagedInstanceRol
eToken: AccessDeniedException: Systems Manager's instance management role is not configured for account: 864981713828
2025-05-10T07:27:36.875815+00:00 ip-172-31-23-124 amazon-ssm-agent.amazon-ssm-agent[947]: 2025-05-10 07:27:36.7230 ERROR
[CredentialRefresher] Retrieve credentials produced error: no valid credentials could be retrieved for ec2 identity. De
fault Host Management Err: error calling RequestManagedInstanceRoleToken: AccessDeniedException: Systems Manager's insta
nce management role is not configured for account: 864981713828
2025-05-16T22:24:44.735197+00:00 ip-172-31-23-124 systemd[1]: apport-autoreport.path - Process error reports when automa
tic reporting is enabled (file watch) was skipped because of an unmet condition check (ConditionPathExists=/var/lib/appo
rt/autoreport).
2025-05-16T22:24:44.735643+00:00 ip-172-31-23-124 systemd[1]: apport-autoreport.timer - Process error reports when autom
atic reporting is enabled (timer based) was skipped because of an unmet condition check (ConditionPathExists=/var/lib/ap
port/autoreport).
```

Explanation:

Searched for all lines containing "error" in the system log.

Task 4.2: Count Errors

Command:

```
grep -c "error" /var/log/syslog
```

```
ubuntu@ip-172-31-23-124:~$ grep -c "error" /var/log/syslog
16
ubuntu@ip-172-31-23-124:~$ |
```

Explanation:

Counted total number of error entries.

Task 4.3: Extract Timestamps & Messages

Command:

```
grep "error" /var/log/syslog | awk '{print $1, $2, $3, $5}'
```

```
ubuntu@ip-172-31-23-124:~$ grep "error" /var/log/syslog | awk '{print $1, $2, $3, $5}'
2025-05-10T06:29:43.849584+00:00 ip-172-31-23-124 systemd[1]: -
2025-05-10T06:29:43.849590+00:00 ip-172-31-23-124 systemd[1]: -
2025-05-10T06:29:50.220892+00:00 ip-172-31-23-124 amazon-ssm-agent.amazon-ssm-agent[947]: 06:29:48.5981
2025-05-10T06:29:50.321227+00:00 ip-172-31-23-124 amazon-ssm-agent.amazon-ssm-agent[947]: 06:29:48.5981
2025-05-10T06:35:16.726805+00:00 ip-172-31-23-124 systemd[1]: -
2025-05-10T06:35:16.729062+00:00 ip-172-31-23-124 systemd[1]: -
2025-05-10T06:59:22.704685+00:00 ip-172-31-23-124 amazon-ssm-agent.amazon-ssm-agent[947]: 06:59:22.6502
2025-05-10T06:59:22.806211+00:00 ip-172-31-23-124 amazon-ssm-agent.amazon-ssm-agent[947]: 06:59:22.6503
2025-05-10T07:27:36.775454+00:00 ip-172-31-23-124 amazon-ssm-agent.amazon-ssm-agent[947]: 07:27:36.7229
2025-05-10T07:27:36.875815+00:00 ip-172-31-23-124 amazon-ssm-agent.amazon-ssm-agent[947]: 07:27:36.7230
2025-05-16T22:24:44.735197+00:00 ip-172-31-23-124 systemd[1]: -
2025-05-16T22:24:44.735643+00:00 ip-172-31-23-124 systemd[1]: -
2025-05-16T22:24:47.866217+00:00 ip-172-31-23-124 amazon-ssm-agent.amazon-ssm-agent[517]: 22:24:47.8657
2025-05-16T22:24:47.966921+00:00 ip-172-31-23-124 amazon-ssm-agent.amazon-ssm-agent[517]: 22:24:47.8657
2025-05-16T22:50:30.071514+00:00 ip-172-31-23-124 amazon-ssm-agent.amazon-ssm-agent[517]: 22:50:30.0150
2025-05-16T22:50:30.172561+00:00 ip-172-31-23-124 amazon-ssm-agent.amazon-ssm-agent[517]: 22:50:30.0150
ubuntu@ip-172-31-23-124:~$ |
```

Explanation:

Extracted timestamp and source of each error log.

Task 4.4: Summarize Unique Error Sources

Command:

```
grep "error" /var/log/syslog | awk '{print $5}' | sort | uniq -c | sort -nr
```

```
ubuntu@ip-172-31-23-124:~$ grep "error" /var/log/syslog | awk '{print $5}' | sort | uniq -c | sort -nr
6 -
2 22:50:30.0150
2 22:24:47.8657
2 06:29:48.5981
1 07:27:36.7230
1 07:27:36.7229
1 06:59:22.6503
1 06:59:22.6502
ubuntu@ip-172-31-23-124:~$ |
```

Explanation:

Summarized error sources and their frequency using a combination of grep, awk, sort, and uniq.

